



Three-dimensional acoustic metamaterial Luneburg lenses for broadband and wide-angle underwater ultrasound imaging

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ABSTRACT

This paper introduces spherical and ellipsoidal metamaterial lenses designed based on the Luneburg lens and transformation optics for underwater ultrasound imaging, which enhances signals over a broadband frequency range and detects objects at a wide-angle. The meta-atoms were analyzed with dispersion curves, equi-frequency contours, and effective acoustic parameters to design the lenses and predict the focusing performance. The three-dimensional lenses made of stainless steel were demonstrated numerically and experimentally in an underwater environment to determine the focal length and full width at half maximum. The spherical lens is omnidirectional. The focal length and the full width at half maximum of the spherical lens are 0λ and 0.6λ at 80 kHz, respectively. The ellipsoidal lens is thinner on the x-axis than that of the spherical lens. This lens has broadband properties by achieving the focal length and the full width at half maximum of 1.5λ to 0λ and 0.95λ to 0.59λ at frequencies between 60 kHz and 160 kHz, respectively. These lenses can potentially be applied to sound navigation ranging or medical transducers, providing the opportunities to view the underwater world with improved resolution at broadband and wide-angle.

1. Introduction

An aberration-free spherical Luneburg lens [1] is a gradient index (GRIN) lens that concentrates propagating waves to the opposite surface of the lens. The lens was modified to generate an inner focus and an outer focus [2], and it was numerically generalized [3]. Although the lens is not perfect [4] because of wavelength limitations [5], it exhibits advantages over conventional lenses because its focus does not suffer from aberrations at all incident angles. The applicability of the lenses in areas such as radar detection and antennas that require wide-angle scanning and high-gain radiation performance [6–12] have been widely researched owing to the excellent focusing, beam steering, and collimation capabilities.

With the advent of metamaterials, extensions into the acoustic realm, and applications of transformation optics (TO) [13–22], the

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